

Basic Probability

1 States of the World

Definition Let S be a set of “states of the world.” Let $\gamma_s > 0$ be the probability of each $s \in S$, so

$$\sum_{s \in S} \gamma_s = 1.$$

The pair (S, γ) is called a probability space.

Given any subset $E \subset S$, we define

$$\gamma(E) \equiv \sum_{s \in E} \gamma_s.$$

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For example, let S be the set of all 36 ordered pairs (i, j) where i and j are integers between 1 and 6. Let $\gamma_s = 1/36$ for each $s \in S$. We might think of S as the possible displays from a roll of two dice.

A picture of the dice state space is given below:

	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

Each state s corresponds to an entry in the above matrix. Let E be the states in which the first die shows 3. Then $E = \{(i, j) : i = 3\} = \{(3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6)\}$. Thus $\gamma(E) = 6/36 = 1/6$.

A random variable X on S assigns a number X_s to each state s . We shall often think of X_s as the payoff from asset X in state s . For example, X_s could be the face of the first die, $X(i, j) \equiv i$ for all $(i, j) \in S$. Similarly, Y_s could be the face of the second die, $Y(i, j) = j$. Notice that a state s determines the outcome of both random variables. The random variable X is constant along each row, while the random variable Y is constant down each column. We can define a third random variable Z on S by $Z_s \equiv X_s + Y_s$ for all $s \in S$. Thus Z is the total rolled by both dice. Notice that Z is constant along all southwest–northwest diagonals of the state space. If $u : \mathbb{R} \rightarrow \mathbb{R}$ is any function on the reals, then $u(X)$ is a random variable if x is, defined by $u(X)_s = u(X_s)$. Again we emphasize that each state s determines the outcomes for all random variables (such as $X, Y, Z, u(X)$) defined on S .

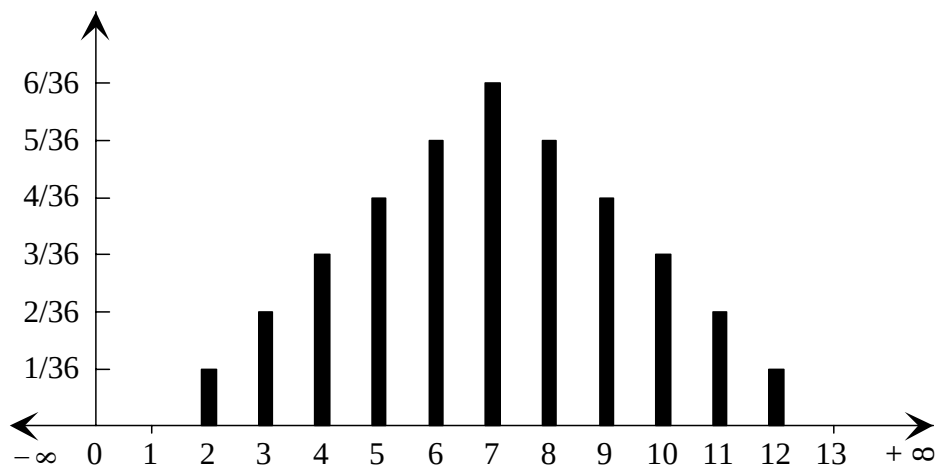
2 Picturing Random Variables

We can explicitly describe how Z depends on the state s by writing the value of Z_s in the cell corresponding to each state s :

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

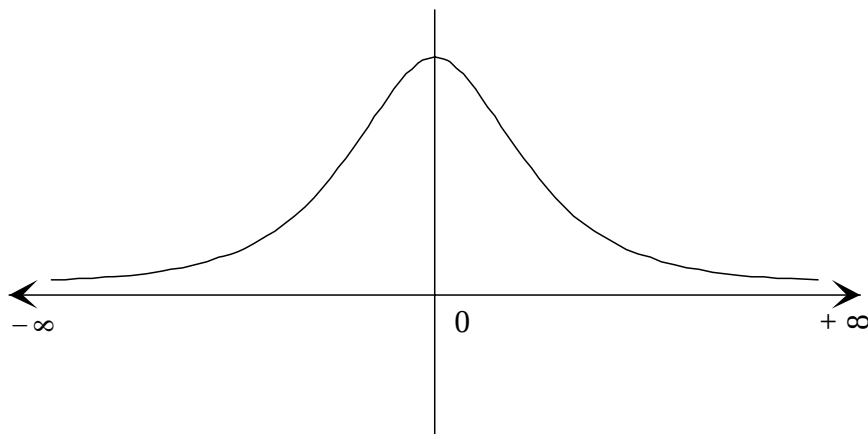
If the state space is large, the enumeration of all values of Z is hard to see. One simplification is to concentrate on the outcome of Z , $z \in \mathcal{R}(Z)$. The *histogram* of Z is the graph associating to each outcome $z \in \mathcal{R}(Z)$ its probability $\gamma(\{s \in S : Z_s = z\})$.

For the sum of two dice, the histogram is:

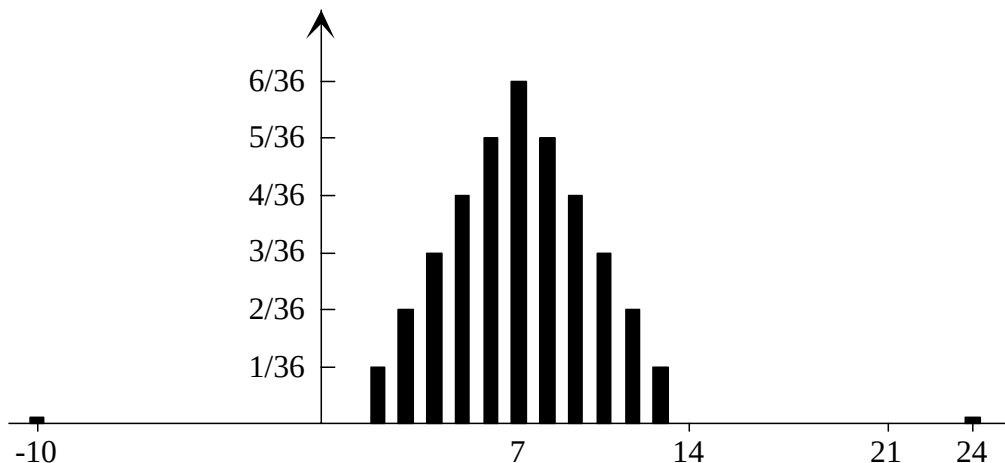


Notice that in this histogram the highest probability is on the central value $z = 7$, and the probability falls quickly to zero as z diverges from the central value.

The most famous histogram is the so-called normal bell curve $f(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2}$ which falls doubly exponentially fast to zero as $|z| \rightarrow \infty$. The normal histogram is said to have a “thin tail.” By contrast, the Cauchy distribution $f(z) = \frac{1}{\pi(1+z^2)}$ has a “fat tail.”



To give an idea of a fat tail, think of a modified dice in which there were two more outcomes, -10 and 24 , each with probability $1/200$. Suppose that probability of $Z = 2, 3, \dots, 12$ was the old probability multiplied by $99/100$. The histogram then would look roughly like



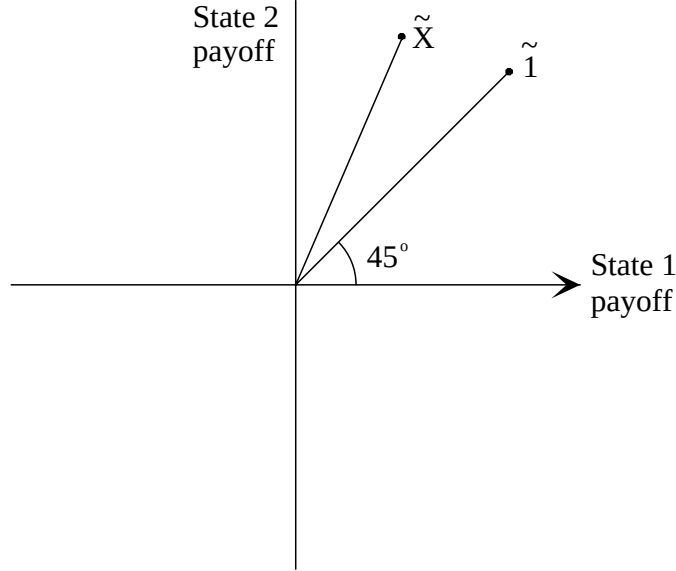
with two points way out in the tails.

A second way of visualizing a random variable X is as a vector. Arbitrarily ordering the states from 1 to S , we can picture the *vector* $X = (X_1, X_2, \dots, X_S)$. For example, in the dice state space, we can number the states proceeding left to right along each row, taking the rows from top to bottom. That gives a vector X in 36-dimensional space.

Once we have represented X as a vector, we can take advantage of geometry to help us understand probabilistic properties of X . The most visually clear notions in Euclidean geometry are the lengths of vectors and the angles between them. (The extreme angle is 90° , when two lines are perpendicular.) Once we rescale the units X_s is measured in, these geometric concepts will correspond to basic probabilistic concepts.

In the geometry of random variables we must not lose track of the probabilities γ_s , $s \in S$. It turns out that the best way of incorporating them is to rescale the units so that X_s is replaced by $X_s \sqrt{\gamma_s}$. Recall that in Euclidean geometry, the length of a vector $(X_1, X_2, \dots, X_S)$ is given by

$\|X\| \equiv \sqrt{\sum_{s=1}^S X_s^2}$. By rescaling the units we get $\|X\|_\gamma \equiv \sqrt{\sum_{s=1}^S (X_s(\sqrt{\gamma_s})^2} = \sqrt{\sum_{s=1}^S \gamma_s X_s^2}$. In particular, if $\tilde{1} \equiv (1, 1, \dots, 1)$ denotes the “riskless” vector of all 1’s, then $\|\tilde{1}\| = S$, while $\|\tilde{1}\|_\gamma = 1$.



3 The Algebra of Random Variables

Random variables can be added, or multiplied, pointwise. Thus if X represents the random payoff to one investment and Y represents the random payoff to another investment, then $Z \equiv X + Y$, defined by $Z_s = X_s + Y_s$, represents the total payoff if both investments are undertaken. Similarly, $Z \equiv XY$, defined by $Z_s = X_s Y_s$, can help us understand how much the investments reinforce each other. If X_s and Y_s are big and positive (or big and both negative) at the same time, then Z will tend to have large values.

Most importantly, random variables can be multiplied by numbers. Thus $Z \equiv tX$ is defined by $Z_s = tX_s$, for each $s \in S$. If X_s represents the total payoff from each dollar of investment, then tX represents the total payoff from $\$t$ of investments.

4 Mean

Definition The mean or expectation of a random variable X on S is

$$\bar{X} \equiv E_\gamma X \equiv \sum_{s \in S} \gamma_s X_s.$$

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Let $\mathcal{R}(X)$ be the range of the random variable X , that is, $x \in \mathcal{R}(X)$ iff $X_s = x$ for some $s \in S$. Then we can also write the expectation of X as

$$E_\gamma(X) = \sum_{x \in \mathcal{R}(X)} \gamma\{s : X(s) = x\}x,$$

which can be quickly read off of the histogram.

For example, to calculate the expectation of the first die, we should go through every state, say row by row, writing

$$\begin{aligned}
E_\gamma(X) &= \underbrace{\frac{1}{36}1 + \frac{1}{36}1 + \cdots + \frac{1}{36}1}_{6 \text{ times}} + \underbrace{\frac{1}{36}2 + \cdots + \frac{1}{36}2}_{6 \text{ times}} \\
&+ \underbrace{\frac{1}{36}3 + \cdots + \frac{1}{36}3}_{6 \text{ times}} + \underbrace{\frac{1}{36}4 + \cdots + \frac{1}{36}4}_{6 \text{ times}} \\
&+ \underbrace{\frac{1}{36}5 + \cdots + \frac{1}{36}5}_{6 \text{ times}} + \underbrace{\frac{1}{36}6 + \cdots + \frac{1}{36}6}_{6 \text{ times}} \\
&= \frac{6}{36}1 + \frac{6}{36}2 + \frac{6}{36}3 + \frac{6}{36}4 + \frac{6}{36}5 + \frac{6}{36}6 \\
&= \frac{1}{6}1 + \frac{1}{6}2 + \frac{1}{6}3 + \frac{1}{6}4 + \frac{1}{6}5 + \frac{1}{6}6 \\
&= 3.5
\end{aligned}$$

By looking at the histogram of X instead, we could have gone directly to the last two lines in the previous derivation. To calculate the expectation of rolling two dice, we can quickly use the histogram of Z and get

$$\begin{aligned}
E_\gamma(Z) &= \frac{1}{36}2 + \frac{2}{36}3 + \frac{3}{36}4 + \frac{4}{36}5 + \frac{5}{36}6 + \frac{6}{36}7 + \frac{5}{36}8 + \frac{4}{36}9 + \frac{3}{36}10 + \frac{2}{36}11 + \frac{1}{36}12 \\
&= \frac{252}{36} = 7
\end{aligned}$$

There is another, geometric way, of visualizing the mean of a random variable X . Consider the vector representation of X , and let $a\tilde{1}$ be the closest point to X on the line $t\tilde{1}$, in the $\sqrt{\gamma_s}$ -weighted Euclidean metric. Then $E_\gamma(X) = a$, as we prove in the next theorem.

Theorem *Let X be an arbitrary random variable. The solution to the problem*

$$\min_a \sum_{s \in S} \gamma_s (X_s - a)^2 \equiv \min_a [\gamma_1 (X_1 - a)^2 + \gamma_2 (X_2 - a)^2 + \cdots + \gamma_s (X_s - a)^2]$$

is uniquely given by $a = E_\gamma(X)$.

Proof If a does minimize the sum, then it must make the derivative 0:

$$\begin{aligned}
0 &= \frac{d}{da} \left(\sum_{s \in S} \gamma_s (X_s - a)^2 \right) \\
&= \sum_{s \in S} 2\gamma_s (X_s - a).
\end{aligned}$$

So

$$2 \sum_{s \in S} \gamma_s a = 2 \sum_{s \in S} \gamma_s X_s$$

Since $\sum_{s \in S} \gamma_s = 1$, we get

$$a = \sum_{s \in S} \gamma_s a = \sum_{s \in S} \gamma_s X_s \equiv E_\gamma(X).$$

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5 Averages of Random Variables

The mean is the most important average of a random variable. But there are others. Let X be a random variable, and let

$$u : \mathbb{R} \rightarrow \mathbb{R}$$

be an increasing function taking the value of the random variable into another value. We can always take the expectation of the random variable $u(X)$. But still more interesting is

$$= CE_u(X) = u^{-1}(E_\gamma[u(X)])$$

which is often called the certainty equivalent of X with respect to u . It is the value x of X such that $u(x)$ is the same as $E_\gamma[u(X)]$. In case u is the identity, the certainty equivalent of any random variable X with respect to u is simply the old mean $E_\gamma[X]$ of X .

But suppose instead that $u(x) = x^2$. If $S = \{1, 2\}$ and $\gamma_1 = 1/2 = \gamma_2$, and $X_s = s$, then

$$\begin{aligned} E_\gamma[u(X)] &= E_\gamma[X^2] = \frac{1}{2}1^2 + \frac{1}{2}2^2 = 2.5 \\ u^{-1}(E_\gamma[u(X)]) &= CE_u(X) = \sqrt{2.5} = 1.6 \end{aligned}$$

Notice that the certainty equivalent of X with respect to u is greater than the mean of X , $E_\gamma[X] = 1.5$.

If we restrict attention to random variables that take on only positive values, then we could consider $u(x) = \sqrt{x} = x^{1/2}$, or $u(x) = \log x$, or $u(x) = -1/x$.

$$\begin{aligned} E_\gamma[X^2] &= \frac{1}{2}1^2 + \frac{1}{2}2^2 = 2.5 \\ CE_{x^2, \gamma}(X) &= \sqrt{2.5} = 1.6 \\ E_\gamma[X] &= \frac{1}{2}1 + \frac{1}{2}2 = 1.5 \\ CE_{x, \gamma}(X) &= 1.5 \\ E_\gamma[\log X] &= \frac{1}{2} \log 1 + \frac{1}{2} \log 2 = \log \sqrt{1 \times 2} = \log 1.4 \\ CE_{\log x, \gamma}(X) &= \exp(\log \sqrt{1 \times 2}) = 1.4 \\ E_\gamma[-X^{-1}] &= \frac{1}{2}(1/1) + \frac{1}{2}(1/2) = -.75 \\ CE_{-x^{-1}, \gamma}(X) &= -(1/(-.75)) = 1.3 \end{aligned}$$

These examples illustrate a famous easy theorem of the Oxford mathematicians Hardy and Littlewood.

Hardy-Littlewood Inequalities: Let X be a random variable, and let $\alpha < \beta$. Then

$$CE_{x^\alpha, \gamma}(X) < CE_{x^\beta, \gamma}(X)$$

In particular,

$$CE_{x^{-1},\gamma}(X) < CE_{\log x,\gamma}(X) < CE_{x,\gamma}(X) < CE_{x^2,\gamma}(X)$$

The logarithmic certainty equivalent is the limit of certainty equivalent of x^α as $\alpha \rightarrow 0$.

Some of these averages have names. $CE_{x^{-1},\gamma}(X) = CE_{-x^{-1},\gamma}(X)$ is called the harmonic mean of X , $CE_{\log x,\gamma}(X) = \times_{s \in S} X_s^{\gamma_s}$ is called the geometric mean of X , and $CE_{x,\gamma}(X) = E_\gamma[X]$ is called the arithmetic mean of X . Thus the Hardy-Littlewood inequalities include as a special case the famous Greek theorem that

Theorem of the Means: For any random variable X , the arithmetic mean is greater than the geometric mean which is greater than the harmonic mean, unless the random variable is constant, in which case all the means are identical.

6 Variance

We now turn to quantifying the “riskiness” of a random variable. This is summarized by how spread out the histogram is. To reduce that to a single number we give the following:

Definition The *variance* of a random variable X is defined by

$$\sigma_X^2 \equiv \text{Var}_\gamma(X) \equiv \sum_{s \in S} \gamma_s (X_s - \bar{X})^2.$$

■

The variance of the die is

$$\begin{aligned} & \frac{6}{36}(1 - 3.5)^2 + \frac{6}{36}(2 - 3.5)^2 + \frac{6}{36}(3 - 3.5)^2 + \frac{6}{36}(4 - 3.5)^2 + \frac{6}{36}(5 - 3.5)^2 + \frac{6}{36}(6 - 3.5)^2 \\ &= \frac{6}{36}(6.25 + 2.25 + .25 + .25 + 2.25 + 6.25) \\ &= \frac{6}{36}(17.50) = 2.916. \end{aligned}$$

Definition The *standard deviation* of a random variable X is defined by

$$\sigma_X \equiv \sqrt{\text{Var}_\gamma(X)} = \sqrt{\sum_{s \in S} \gamma_s (X_s - \bar{X})^2}.$$

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The standard deviation of the die is thus $\sigma_X \equiv \sqrt{2.916} \approx 1.7$.

A random variable with 0 variance has 0 standard deviation, and always turns up equal to its expectation. The variance is a measure of the average squared deviation from the mean. From the Hardy-Littlewood inequalities, we know that the standard deviation is always at least as big as the average absolute deviation from the mean:

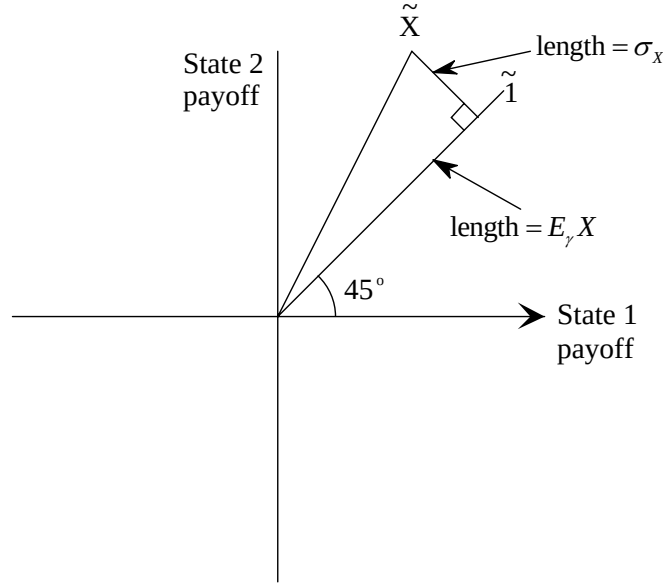
$$\sqrt{\sum_{s \in S} \gamma_s (X_s - \bar{X})^2} \geq \sum_{s \in S} \gamma_s |X_s - \bar{X}|.$$

For example, with the die, $\sigma_X = \sqrt{2.916} \approx 1.7$. The average absolute deviation is

$$\frac{6}{36}(2.5 + 1.5 + .5 + .5 + 1.5 + 2.5) = 1.5.$$

The reason for the fame of the standard deviation (as opposed to expected absolute deviation) has to do with the computational and geometric simplicity of variance, and the fact that normally distributed random variables are defined by their mean and standard deviation.

We can visualize the standard deviation of a random variable X by noting that it is the (γ -Euclidean) distance from X to $[E_\gamma(X)]\tilde{1}$.



This geometric decomposition of any vector X into its mean and standard deviation is another reason that those two concepts have been singled out as special in probability theory. We write this formally as:

Theorem For any random variable X ,

$$E_\gamma(X^2) \equiv \sum_{s \in S} \gamma_s X_s^2 = \sum_{s \in S} \gamma_s \bar{X}^2 + \sum_{s \in S} \gamma_s (X_s - \bar{X})^2 \equiv [E_\gamma(X)]^2 + \text{Var}_\gamma(X).$$

Proof

$$\begin{aligned} \sum_{s \in S} \gamma_s (X_s - \bar{X})^2 &= \sum_{s \in S} \gamma_s X_s^2 - 2 \sum_{s \in S} \gamma_s X_s \bar{X} + \sum_{s \in S} \gamma_s \bar{X}^2 \\ &= \sum_{s \in S} \gamma_s X_s^2 - 2 \bar{X}^2 + \bar{X}^2 \\ &= \sum_{s \in S} \gamma_s X_s^2 - \bar{X}^2 \end{aligned}$$

■

The variance does quantify how risky a random variable is in the following sense. People have defined the “value at risk” as the maximum amount by which X could fall short of its expectation $E_\gamma(X)$ with probability at least 5%. In other words, if $\gamma(\{s \in S : X_s \leq E_\gamma(X) - L\}) \geq .05$, then the value at risk is at least L . The variance puts a bound on the value at risk.

Theorem

$$L \leq \frac{\sigma_X}{\sqrt{\gamma(\{s \in S : X_s \leq E_\gamma(X) - L\})}}.$$

In particular,

$$\text{Value at risk} \leq \sigma_X \sqrt{20}.$$

Proof

$$\begin{aligned} \sigma_X^2 &\geq L^2 \gamma(\{s \in S : (X_s - E_\gamma(X))^2 \geq L^2\}) \\ &\geq L^2 \gamma(\{s \in S : X_s - E_\gamma(X) \leq -L\}). \end{aligned}$$

■

This bound can be made even tighter for normally distributed random variables. For normally distributed random variables, a loss of $2\sigma_X$ (a “two standard deviation loss”) has probability less than 2.5%.

The following property is important for variance.

Theorem For any random variable X , and constants t and a , $\text{Var}(tX + a\mathbf{1}) = t^2 \text{Var}(X)$.

Proof

$$\begin{aligned} &\sum_{s \in S} \gamma_s (tX_s + a - [t\bar{X} + a])^2 \\ &= \sum_{s \in S} \gamma_s (tX_s - t\bar{X})^2 \\ &= t^2 \sum_{s \in S} \gamma_s (X_s - \bar{X})^2. \end{aligned}$$

■

We now consider two or more random variables defined on the same state space S .

Theorem Let X , Y and Z be random variables on S with $Z = X + Y$. Then

$$E_\gamma(Z) = E_\gamma(X) + E_\gamma(Y).$$

Proof

$$\sum_{s \in S} \gamma_s (X_s + Y_s) = \sum_{s \in S} \gamma_s X_s + \sum_{s \in S} \gamma_s Y_s.$$

■

Clearly if $Z = tX$ for such number t , then $E_\gamma(Z) = tE_\gamma(X)$. Thus expectation is a linear operator on random variables.

In our example, let Y be the outcome of the second die $Y(i, j) = j$. Let Z be the sum showing on both die. Using the last theorem,

$$EZ = 7 = 3.5 + 3.5 = E(X) + E(Y).$$

Note how much simpler this is than calculating the expectation from the histogram..

7 Covariance

Definition The covariance of two random variables X and Y is:

$$\sigma_{XY} = \text{Cov}_\gamma(X, Y) \equiv \sum_{s \in S} \gamma_s(X_s - \bar{X})(Y_s - \bar{Y}).$$

Clearly $\text{Var}_\gamma(X) = \text{Cov}_\gamma(X, X)$ for any random variable X . If $\text{Cov}_\gamma(X, Y) > 0$, the two variables tend to “move together”: when X_s is above its mean, Y_s is usually above its mean. Clearly $\sigma_{XY} = \sigma_{YX}$.

Lemma If $Y \equiv a$ is a constant, then $\text{Cov}_\gamma(X, Y) = 0$, for all random variables X .

Proof

$$\text{Cov}_\gamma(X, Y) = \sum_{s \in S} \gamma_s(X_s - \bar{X})(a - a) = 0.$$

Lemma Let X, Y, Z be random variables. Then

$$\text{Cov}_\gamma(X, Y + Z) = \text{Cov}_\gamma(X, Y) + \text{Cov}_\gamma(X, Z).$$

Proof

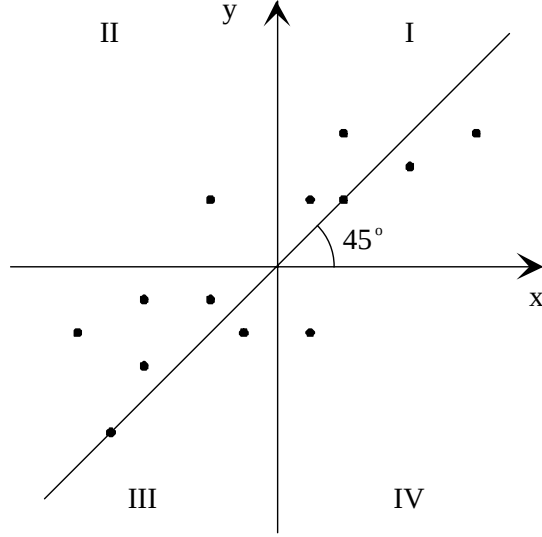
$$\begin{aligned} & \sum_{s \in S} \gamma_s(X_s - \bar{X})(Y_s + Z_s - (\bar{Y} + \bar{Z})) \\ &= \sum_{s \in S} \gamma_s(X_s - \bar{X})(Y_s - \bar{Y} + Z_s - \bar{Z}) \\ &= \sum_{s \in S} \gamma_s(X_s - \bar{X})(Y_s - \bar{Y}) + \sum_{s \in S} \gamma_s(X_s - \bar{X})(Z_s - \bar{Z}). \end{aligned}$$

Clearly $\text{Cov}_\gamma(X, tY) = t\text{Cov}_\gamma(X, Y)$ for any number t . Hence, fixing X , $\text{Cov}_\gamma(X, Y)$ is a linear operator on random variables Y .

Definition If $\sigma_X > 0$ and $\sigma_Y > 0$, the correlation of X and Y is defined by $\rho_{XY} \equiv \sigma_{XY} / \sigma_X \sigma_Y$.

We can give two geometric interpretations of the correlation of random variables X and Y . First, let us put them in their so-called “standard form.” Define the random variables x and y by $x_s \equiv (X_s - \bar{X}) / \sigma_X$ and $y_s \equiv (Y_s - \bar{Y}) / \sigma_Y$. This simply amounts to changing the scale in which the variables are measured. Note that $E_\gamma(x) = E_\gamma(y) = 0$ and $\sigma_x = \sigma_y = 1$. Note also that $\rho_{XY} = \sigma_{xy}$. The correlation of the random variables X and Y is identical to the covariance between the standardized random variables x and y . To understand when X and Y are positively correlated or uncorrelated, or negatively correlated, it suffices to understand the correlation of x and y .

In our first geometric interpretation, we can graph all the ordered pairs (x_s, y_s) for every $s \in S$.



Since $\sigma_{xy} \equiv \sum_{s \in S} \gamma_s x_s y_s$, to the extent the points (x_s, y_s) are simultaneously positive (in the first quadrant) or simultaneously negative (in the third quadrant) the correlation is positive. Indeed, if the points were all on the 45° line $y = x$, we would say that x and y (and therefore X and Y) are perfectly correlated. In that case $\sigma_{xy} \equiv \sum_{s \in S} \gamma_s x_s y_s = \sum_{s \in S} \gamma_s x_s^2 = \sigma_x^2 = 1$.

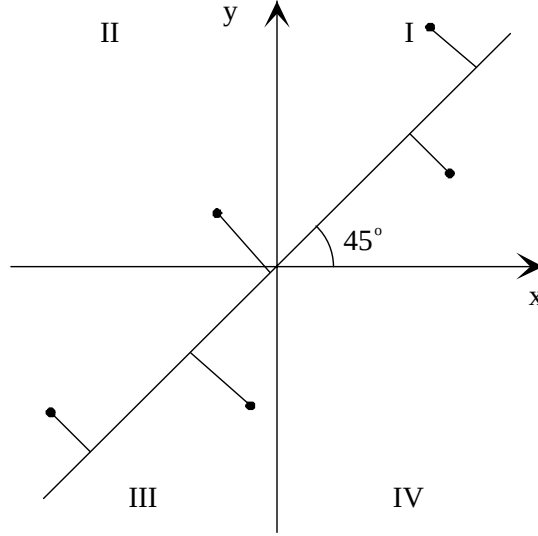
If most of the points (x_s, y_s) are in the 2nd and 4th quadrants, the correlation of x and y is negative (when one goes up, the other tends to go down). Indeed, if the points were all on the line $y = -x$, we would say that x and y (and therefore X and Y) are perfectly negatively correlated. In that case $\sigma_{xy} \equiv \sum_{s \in S} \gamma_s x_s y_s = \sum_{s \in S} -\gamma_s x_s^2 = -\sigma_x^2 = -1$.

It only remains to interpret the correlation in the in-between cases where the points (x_s, y_s) are not on either diagonal. To do this, note first that the closest point to (x_s, y_s) on the line $y = x$ is $((x_s + y_s)/2, (x_s + y_s)/2)$. The difference $(x_s, y_s) - ((x_s + y_s)/2, (x_s + y_s)/2) = ((x_s - y_s)/2, (y_s - x_s)/2)$ is clearly orthogonal (in the standard Euclidean sense) to the line $y = x$, proving that $((x_s + y_s)/2, (x_s + y_s)/2)$ is indeed the closest point on $y = x$ to (x_s, y_s) .

Next, let us square this distance: $(x_s - y_s)^2/4 + (y_s - x_s)^2/4 = (2x_s^2 + 2y_s^2 - 4x_s y_s)/4$. Adding over all $s \in S$, weighted by the probabilities γ_s , we get that the average squared difference d^2 of (x_s, y_s) from the line $y = x$ is

$$d^2 \equiv \sum_{s \in S} \gamma_s \left(\frac{2x_s^2 + 2y_s^2 - 4x_s y_s}{4} \right) = \frac{1}{2} + \frac{1}{2} - \sum_{s \in S} \gamma_s x_s y_s = 1 - \sigma_{xy}.$$

Since d^2 is the sum of nonnegative numbers, $d^2 \geq 0$ and so $\sigma_{xy} \leq 1$. (By replacing x with $-x$, the same argument shows that $-\sigma_{xy} \leq 1$.) Moreover, we can interpret $\sigma_{xy} \equiv 1 - d^2$. Since $d^2 > 0$ (unless $x_s = y_s$ for all s), we have that $-1 < \sigma_{xy} < 1$ unless x and y are perfectly correlated or perfectly negatively correlated.



Summarizing our results so far we have:

Theorem *Let X and Y be arbitrary random variables with $\sigma_X > 0$ and $\sigma_Y > 0$. Let x and y be their standard versions. Then $|\rho_{XY}| \equiv |\sigma_{XY}/\sigma_X\sigma_Y| \equiv |\sigma_{xy}| \leq 1$, with equality only if X and Y are perfectly correlated or perfectly negatively correlated.*

We can also give a second interpretation of the correlation $\rho_{XY} = \sigma_{xy}$. Suppose we must find the line $y = ax$ that best predicts y , given any value of x . What should a be? The answer depends on the punishment criterion for guessing wrong. But if the punishment is the square of the error, then minimizing the expected punishment requires putting $a = \sigma_{xy}$.

Theorem Let x and y be arbitrary random variables with $\sigma_x^2 = 1$. Then the solution to

$$\min_a \sum_{s \in S} \gamma_s (y_s - ax_s)^2$$

occurs at $a = \sum_{s \in S} \gamma_s x_s y_s$.

Proof

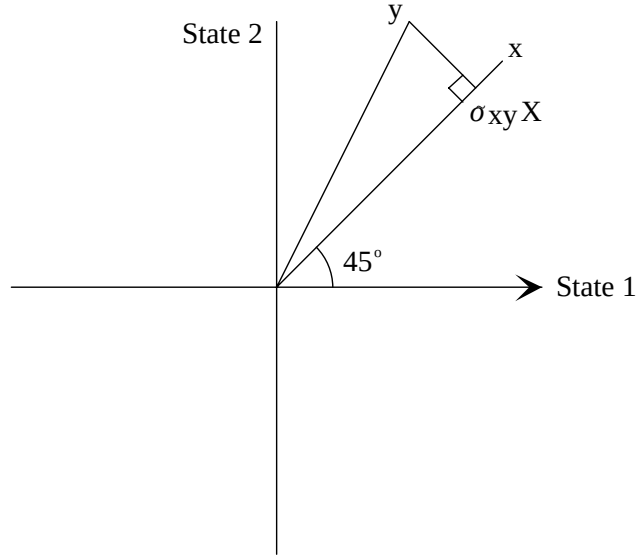
$$\begin{aligned} 0 &= \frac{d}{da} \left(\sum_{s \in S} \gamma_s (y_s - ax_s)^2 \right) \\ &= 2 \sum_{s \in S} \gamma_s (y_s - ax_s) x_s \end{aligned}$$

So

$$\sum_{s \in S} \gamma_s x_s y_s = a \sum_{s \in S} \gamma_s x_s^2 = a.$$

■

Geometrically, we can picture this by thinking of x and y as vectors in $\mathbb{R}^S$.



Definition The random variables X and Y are called uncorrelated if $\text{Cov}_\gamma(X, Y) = 0$.

8 Independence

Definition The random variables X and Y are independent if for all numbers a and b ,

$$\sum_{s \in X^{-1}(a) \cap Y^{-1}(b)} \gamma_s \equiv \text{Prob}(X = a \& Y = b) = [\text{Prob}(X = a)][\text{Prob}(Y = b)].$$

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The random variables are independent if and only if knowing which value Y takes on provides no information about which value X takes on and vice versa. The best guess of Y_s given X_s is $E_\gamma Y$. Hence from the second interpretation of correlation we get:

Theorem If X and Y are independent, then $\text{Cov}_\gamma(X, Y) = 0$.

Theorem If X and Y are independent, then

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Proof

$$\begin{aligned} \text{Var}_\gamma(X + Y) &= \text{Cov}_\gamma(X + Y, X + Y) \\ &= \text{Cov}_\gamma(X + Y, X) + \text{Cov}_\gamma(X + Y, Y) \\ &= \text{Cov}_\gamma(X, X) + \text{Cov}_\gamma(Y, X) + \text{Cov}_\gamma(X, Y) + \text{Cov}_\gamma(Y, Y) \\ &= \text{Cov}_\gamma(X, X) + 0 + 0 + \text{Cov}_\gamma(Y, Y). \end{aligned}$$

■

In the dice example, X and Y are clearly independent. The variance of each die was calculated at 2.916. Hence the variance of their sum Z must be

$$\text{Var}_\gamma(Z) = \text{Cov}_\gamma(X + Y, X + Y) = \text{Var}_\gamma(X) + \text{Var}_\gamma(Y) = 5.833.$$

One can verify this by computing

$$\begin{aligned} \text{Var}_\gamma(Z) &= \frac{1}{36}(2-7)^2 + \frac{2}{36}(3-7)^2 + \frac{3}{36}(4-7)^2 + \frac{4}{36}(5-7)^2 + \frac{5}{36}(6-7)^2 + \frac{6}{36}(7-7)^2 \\ &\quad + \frac{5}{36}(8-7)^2 + \frac{4}{36}(9-7)^2 + \frac{3}{36}(10-7)^2 + \frac{2}{36}(11-7)^2 + \frac{1}{36}(12-7)^2 \\ &= \frac{25 + 32 + 27 + 16 + 5 + 0 + 5 + 16 + 27 + 32 + 25}{36} = \frac{210}{36}. \end{aligned}$$

If X represents the change in stock prices over the first day, and Y represents the subsequent change over the second day, then the two-day change Z has an expected change equal to the sum of each day's expected changes. If the changes are independent, then the 2-day variance is the sum of the 1-day variances. More generally, if daily fluctuations are independent, then annual expected price changes are the sum of daily expected changes, and annual variances are the sum of daily variances. (Thus both annual numbers are around 250 times bigger than the daily numbers.)

In our dice example, X and Y are clearly independent. So

$$\begin{aligned} \text{Cov}_\gamma(X, Z) &= \text{Cov}_\gamma(X, X + Y) = \text{Cov}_\gamma(X, X) + \text{Cov}_\gamma(X, Y) \\ &= \text{Var}_\gamma(X) + 0 = 2.916. \end{aligned}$$

Thus

$$\rho_{XZ} \equiv \frac{\sigma_{XZ}}{\sigma_X \sigma_Z} = \frac{2.916}{\sqrt{2.916} \sqrt{2(2.916)}} = \frac{1}{\sqrt{2}} \approx .7.$$

9 Estimating Mean and Variance

Suppose we have a random variable X , but we do not know the values X_s that it takes on. Suppose however that we can *sample* from S . Let $\bar{S}$ be a state space generated by repeatedly drawing a random state $s \in S$ according to the probability γ . For each s drawn, we observe the values of all random variables at s . The implicit probability $\bar{\gamma}$ on $\bar{S}$ is $\bar{\gamma}_{\bar{s}} = 1/\#\bar{S}$ for all $\bar{s} \in \bar{S}$.

To what extent is $\bar{S}$ a faithful representation of S ? Clearly if $\#\bar{S} < \#S$, important possibilities might be left out. But what if $\#\bar{S} \gg \#S$? Chances are, the number of times a state s is chosen as an $\bar{s} \in \bar{S}$ will be roughly proportional to γ_s once $\#\bar{S}$ becomes very large. Of course, if γ_s is an irrational number then the proportion of $\bar{s} = s$ can never be exactly γ_s . But it could converge as $\#\bar{S} \rightarrow \infty$. The question is, how fast? The answer is, slower than you might think.

For example, suppose that S consists of two points, s_1 and s_2 . Suppose $\gamma_1 = 999/1000$ and $\gamma_2 = 1/1000$. If we draw a sample of 1000 points, the chances are a little more than 36% that every $\bar{s} = s_1$, and s_2 never appears. The odds are exactly $(999/1000)^{1000} = (1 - (1/1000))^{1000} \approx 1/e \approx 36\%$. If 2000 points are drawn, the odds are still about $1/e^2 \approx 13\%$ that s_2 never appears. If s_2 represents a catastrophe for some random variable, then it is not so unlikely that the sample space $\bar{S}$ will not reveal the real danger.